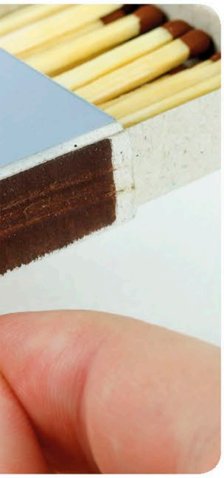


Activation energy

Some chemical reactions need energy to start them off. The heat energy made when striking a match starts a reaction in the chemicals in the match tip and the surface of the box that produces a flame.

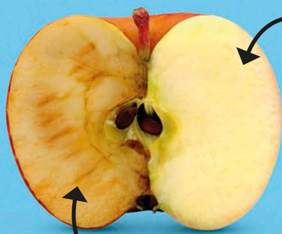
The friction of striking a match provides the activation energy to make a flame.

The foam expands and moves out of the glass.

Forming foam
The energy released during the reaction produces lots of heat and the movement of the oxygen. The large amount of foam spills out of the glass like toothpaste, which gives the reaction its name.

Everyday reactions

Chemical reactions can release lots of energy when chemicals bubble or explode, but there are much slower, less noticeable chemical reactions happening all around us.



Brown apple
The inside of a freshly cut apple reacts with oxygen in the air in a chemical reaction that makes the flesh turn brown.

Rusting nail
Iron and steel slowly react with water and oxygen in the air to form a new substance, called rust.

The colourless part of the egg turns white when it is cooked.

Frying an egg
When you heat up an egg, the proteins inside it react and join up with each other. The joined-up proteins make the egg turn white.